

1. Raw data (Feng et al., 2019, NASDAQ)

1,026 tickers $\times$ 1,246 trading days;
industry relation tensor ($1,026 \times 1,026 \times 97$)

2. Preprocessing

Treat the -1234 sentinel as missing
Features: normalized close + 5-, 10-, 20-, 30-day moving averages
Target: next-day return ratio

3. Split by date (same for both models)

train: days 0-755 | validation: 756-1007 | test: 1008-

4a. Baseline: Rank_LSTM

LSTM over each stock's own
history; no relation graph

4b. RSR (implicit, RSR_I)

baseline's exported embedding
+ attention over the
static industry graph

5. Training (identical settings, two independent runs)

Adam, seq = 4, unit = 32; loss = MSE + $\alpha \times$ pairwise ranking penalty
50 epochs; test metrics taken at the best-validation epoch

6. Evaluation on the test split

MSE | MRR (reciprocal true rank of top pick) | IRR (return of top pick)
compared with Table 6 (NASDAQ) of the original paper